NANO 2314 Quiz 09 — Review Packet

Practice for Quiz 09, Friday, November 6, 2026 · Fall 2026 · Prof./Coach Emori

15 free-response problems on the ideas Quiz 09 covers, from two groups up to three groups with a missing arrow. Write your answer, with your work, in the space given. Try Part 1 first, then check yourself in Part 2. Multi-step answers are drawn step by step, and the **In words** box under each is the kind of written answer that earns full credit. No calculator needed.

The **Quiz 09 Practice Drill** (https://claude.ai/artifact/SGQnbFtABUnMAhqjz7KEaq) gives fresh versions of all of these ideas, with new numbers each round.

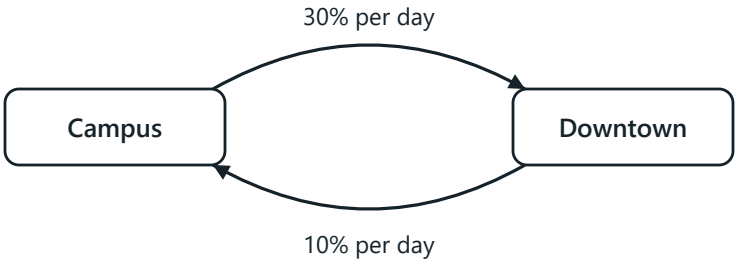
Topic	Practices	Problems
1. Step-by-step equations	from a diagram of rates to one equation per group, and back	1–4
2. The 3×3 matrix	rows are "to," columns are "from," each column adds up to 1	5–7
3. The eigenvalue equation and λ	rewriting the equation, how many eigenvalues, why λ = 1	8–9
4. Verify by multiplying	matrix times vector by hand, and what the result shows	10–13
5. Physical meaning	what the settled numbers tell you, and what keeps moving	14–15

Part 1 · Problems

1. Step-by-step equations

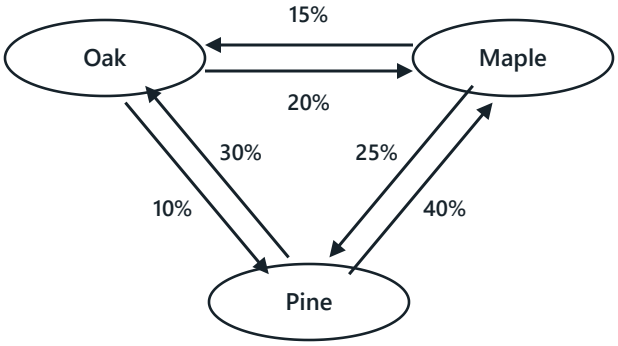
Problem 1 Warm-up: two groups

Every day, bikes move between two stations as shown, and the total stays the same. Let c_n and d_n be the numbers of bikes at Campus and Downtown on day n. Write a system of two equations that gives c_n and d_n in terms of c_{n-1} and d_{n-1}.



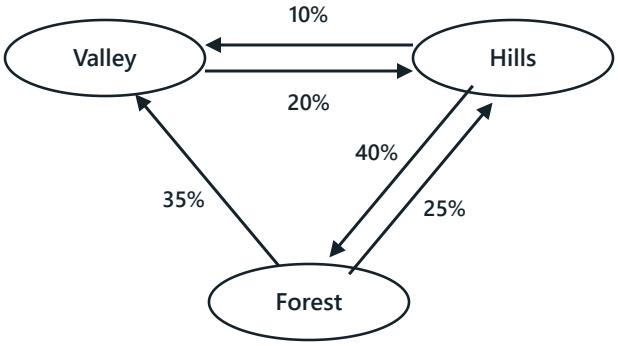
Problem 2 Three groups, six arrows

Every week, students move between three dining halls as shown, and the total stays the same. Let o_n, m_n, and p_n be the numbers of students at Oak, Maple, and Pine in week n. Write a system of three equations.



Problem 3 Three groups, a missing arrow

Every year, deer move between three habitats as shown, and the total stays the same. Let v_n, h_n, and f_n be the numbers of deer in Valley, Hills, and Forest in year n. Write a system of three equations. The later problems use this model.

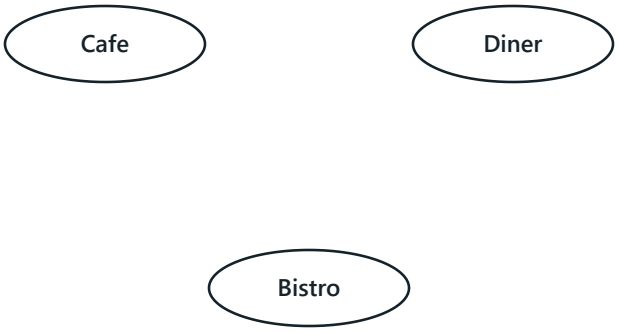


Problem 4 Draw the diagram for a system

Customers move between three restaurants every week, and c_n, d_n, and b_n are the numbers at Cafe, Diner, and Bistro:

$$c_n = 0.65\,c_{n-1} + 0.20\,d_{n-1} + 0.45\,b_{n-1}$$
$$d_n = 0.10\,c_{n-1} + 0.75\,d_{n-1}$$
$$b_n = 0.25\,c_{n-1} + 0.05\,d_{n-1} + 0.55\,b_{n-1}$$

Draw the arrows on the diagram, each with its rate.



2. The 3×3 matrix

Problem 5 From a system to a matrix

Write each system as a matrix equation, with the old numbers on the left and the new numbers on the right.

a. The bikes of Problem 1.

b. The dining halls of Problem 2.

c. The deer of Problem 3.

Problem 6 Rows go to, columns come from

Here is the matrix for the deer, with the groups in the order Valley, Hills, Forest:

$$\begin{pmatrix} 0.80 & 0.10 & 0.35 \\ 0.20 & 0.50 & 0.25 \\ 0 & 0.40 & 0.40 \end{pmatrix}$$

a. Which entry is the rate from Hills to Forest? Give its row, column, and value.

b. What does 0.35 in row 1, column 3 mean?

c. What does 0.50 in row 2, column 2 mean?

d. Why is the entry in row 3, column 1 a 0?

Problem 7 Spot the mistake

A student writes this matrix for the deer:

$$\begin{pmatrix} 0.80 & 0.20 & 0 \\ 0.10 & 0.50 & 0.40 \\ 0.35 & 0.25 & 0.40 \end{pmatrix}$$

a. Add up each column. What do you get?

b. What should each column add up to, and why?

c. What did the student do wrong?

3. The eigenvalue equation and λ

Problem 8 Rewrite as an eigenvalue equation

Rewrite the matrix equation for the deer,

$$\begin{pmatrix} 0.80 & 0.10 & 0.35 \\ 0.20 & 0.50 & 0.25 \\ 0 & 0.40 & 0.40 \end{pmatrix} \begin{pmatrix} v_{n-1} \\ h_{n-1} \\ f_{n-1} \end{pmatrix} = \begin{pmatrix} v_n \\ h_n \\ f_n \end{pmatrix},$$

as an eigenvalue equation, using v, h, f, and λ. What changed?

Problem 9 How many eigenvalues, and which one?

a. How many eigenvalues does a 3×3 matrix have, counting repeats? A 5×5 matrix?

b. The deer matrix has eigenvalues 1, 0.5, and 0.2. Which one is reasonable for the numbers the herd settles at? Explain why.

4. Verify by multiplying

Problem 10 Multiply a matrix by a vector

Multiply by hand, one row at a time.

a. $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 0 & 4 & 1 \end{pmatrix}\begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$

b. $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 2 & 0 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$

c. $\begin{pmatrix} 0.80 & 0.10 & 0.35 \\ 0.20 & 0.50 & 0.25 \\ 0 & 0.40 & 0.40 \end{pmatrix}\begin{pmatrix} 300 \\ 300 \\ 400 \end{pmatrix}$

Problem 11 Verify an eigenvector

Multiply by hand to verify that $(800, 800, 400)$ is an eigenvector of the dining-hall matrix with $\lambda = 1$. Show all three rows.

$$\begin{pmatrix} 0.70 & 0.15 & 0.30 \\ 0.20 & 0.60 & 0.40 \\ 0.10 & 0.25 & 0.30 \end{pmatrix}\begin{pmatrix} 800 \\ 800 \\ 400 \end{pmatrix}$$

Problem 12 Multiply to find λ

The vector $(-50, 10, 40)$ is an eigenvector of the deer matrix. Multiply to find its eigenvalue.

$$\begin{pmatrix} 0.80 & 0.10 & 0.35 \\ 0.20 & 0.50 & 0.25 \\ 0 & 0.40 & 0.40 \end{pmatrix} \begin{pmatrix} -50 \\ 10 \\ 40 \end{pmatrix}$$

Problem 13 Not every vector is an eigenvector

a. Multiply the dining-hall matrix by $(1000, 600, 400)$.

$$\begin{pmatrix} 0.70 & 0.15 & 0.30 \\ 0.20 & 0.60 & 0.40 \\ 0.10 & 0.25 & 0.30 \end{pmatrix} \begin{pmatrix} 1000 \\ 600 \\ 400 \end{pmatrix}$$

b. Is $(1000, 600, 400)$ an eigenvector? How can you tell?

c. Another student says, "Yes, because the total stayed 2000." What would you tell them?

5. Physical meaning

Problem 14 What the eigenvector means

A solver gives the eigenvector $(5, 3, 2)$ with $\lambda = 1$ for the deer, in the order Valley, Hills, Forest. The herd has 1000 deer.

a. What are the settled numbers?

b. In two sentences, say what this vector physically tells you.

Problem 15 Still moving at equilibrium

The deer have settled at $(500, 300, 200)$.

a. How many deer move along each of the five arrows in one year? (See the diagram in Problem 3.)

b. For each habitat, add up what leaves and what arrives.

c. Have the deer stopped moving?

Part 2 · Answers and explanations

1. Step-by-step equations

Problem 1 Warm-up: two groups

$0.7\,c_{n-1}$ ↓ **stay** at Campus $(1 - 0.3)$ + plus $0.1\,d_{n-1}$ ↓ **arrive** from Downtown

↓ add them

$$c_n = 0.7\,c_{n-1} + 0.1\,d_{n-1}$$

$$d_n = 0.3\,c_{n-1} + 0.9\,d_{n-1}$$

In words. Each new number is the bikes that stay (one minus the rate leaving) plus the bikes that arrive from the other station (that station's rate times its number the day before).

Common mistakes

- Swapping the rates: 30% leaves Campus (the arrow points from Campus), and 10% leaves Downtown.
- Using the rate leaving for the bikes that stay ($0.3\,c_{n-1}$ in the Campus equation): those that stay are $1 - 0.3 = 0.7$.

Problem 2 Three groups, six arrows

Leaving Oak: 20% + 10% = 30%. Staying: 70%.

↓ add the arrivals: 15% of Maple and 30% of Pine

$$o_n = 0.70\,o_{n-1} + 0.15\,m_{n-1} + 0.30\,p_{n-1}$$

↓ the same for the other two

$$m_n = 0.20\,o_{n-1} + 0.60\,m_{n-1} + 0.40\,p_{n-1}$$

$$p_n = 0.10\,o_{n-1} + 0.25\,m_{n-1} + 0.30\,p_{n-1}$$

Why: check: the coefficients of o_{n-1} add up to $0.70 + 0.20 + 0.10 = 1$, and the same for m_{n-1} and p_{n-1}.

In words. Each group's new number is the fraction it keeps (one minus every rate leaving it) plus each incoming rate times the group that rate comes from.

Common mistakes

- Subtracting only one rate: two arrows leave Oak, so it keeps $1 - 0.20 - 0.10 = 0.70$, not 0.80 or 0.90.
- Using the arrows leaving a group as its arrivals: in Oak's equation, the arrivals are the arrows pointing into Oak.
- Writing n on the right-hand side: the right-hand side is the week before, with subscript $n - 1$.

Problem 3 Three groups, a missing arrow

Staying: Valley $1 - 0.20 = 0.80$; Hills $1 - 0.10 - 0.40 = 0.50$; Forest $1 - 0.35 - 0.25 = 0.40$

↓ add the arrivals along the arrows into each habitat

$$v_n = 0.80\, v_{n-1} + 0.10\, h_{n-1} + 0.35\, f_{n-1}$$

$$h_n = 0.20\, v_{n-1} + 0.50\, h_{n-1} + 0.25\, f_{n-1}$$

$$f_n = 0.40\, h_{n-1} + 0.40\, f_{n-1}$$

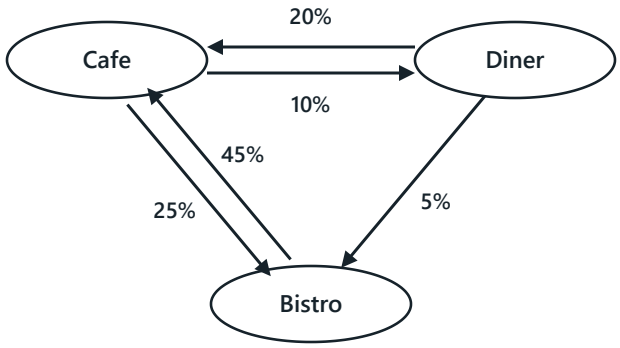
In words. There is no arrow from Valley to Forest, so Forest's equation has no v_{n-1} term. Only one arrow leaves Valley, so Valley subtracts only one rate; two arrows leave Hills and two leave Forest.

Common mistakes

- Giving every habitat two rates to subtract: only the 20% arrow leaves Valley.
- Filling the empty place in Forest's equation with some other rate: no arrow means no term (a 0).

Problem 4 Draw the diagram for a system

A term in one group's equation that multiplies another group's number is an arrow from that other group. In Diner's equation, $0.10\, c_{n-1}$ is 10% from Cafe to Diner. There is no b_{n-1} term there, so no arrow goes from Bistro to Diner. The terms $0.65\, c_{n-1}$, $0.75\, d_{n-1}$, and $0.55\, b_{n-1}$ are the fractions that stay, so they are not arrows.



Common mistakes

- Drawing arrows of 65%, 75%, and 55%: those are the fractions that stay.
- Reversing an arrow: $0.20\, d_{n-1}$ in Cafe's equation multiplies Diner's number, so it goes from Diner to Cafe.

2. The 3×3 matrix

Problem 5 From a system to a matrix

a. $\begin{pmatrix} 0.7 & 0.1 \\ 0.3 & 0.9 \end{pmatrix}\begin{pmatrix} c_{n-1} \\ d_{n-1} \end{pmatrix} = \begin{pmatrix} c_n \\ d_n \end{pmatrix}$

b. $\begin{pmatrix} 0.70 & 0.15 & 0.30 \\ 0.20 & 0.60 & 0.40 \\ 0.10 & 0.25 & 0.30 \end{pmatrix}\begin{pmatrix} o_{n-1} \\ m_{n-1} \\ p_{n-1} \end{pmatrix} = \begin{pmatrix} o_n \\ m_n \\ p_n \end{pmatrix}$

c. The deer, step by step:

Each equation is one row, with the same order of groups on every row.

↓ a missing term is a 0 in its own position

$$\begin{pmatrix} 0.80 & 0.10 & 0.35 \\ 0.20 & 0.50 & 0.25 \\ 0 & 0.40 & 0.40 \end{pmatrix}\begin{pmatrix} v_{n-1} \\ h_{n-1} \\ f_{n-1} \end{pmatrix} = \begin{pmatrix} v_n \\ h_n \\ f_n \end{pmatrix}$$

In words. A row is the group they go to, and a column is the group they come from. Row 3, column 1 is 0 because no arrow goes from Valley to Forest.

Common mistakes

- Sliding the entries of a short equation to the left: 0.40 and 0.40 belong in columns 2 and 3 of row 3, after a 0.
- Putting the vector with subscript n on the left: the matrix acts on the old numbers and gives the new ones.

Problem 6 Rows go to, columns come from

A matrix entry is **row to, column from**.

a. Row 3, column 2: 0.40. Forest is the third group (row), and Hills is the second (column).

b. 35% of the deer in Forest move to Valley every year.

c. 50% of the deer in Hills stay in Hills every year.

d. No deer move from Valley to Forest: there is no such arrow.

Common mistakes

- Reading row 1, column 3 as Valley to Forest: the row is where they go, and the column is where they come from.
- "0.35 of all the deer": an entry is a fraction of the group in its column, not of the total.

Problem 7 Spot the mistake

a. 1.25, 0.95, and 0.80.

b. Each should add up to 1. A column lists where everyone in one group ends up, so its fractions must add up to the whole group. That is why the total is conserved.

c. The rows and columns are swapped: each row was filled with the rates *leaving* a group. Here the rows add up to 1, which is a sign of this mistake.

Common mistakes

- Checking the rows instead of the columns: here the rows add up to 1, but the rows are not what must add up to 1.

3. The eigenvalue equation and λ

Problem 8 Rewrite as an eigenvalue equation

$$\begin{pmatrix} 0.80 & 0.10 & 0.35 \\ 0.20 & 0.50 & 0.25 \\ 0 & 0.40 & 0.40 \end{pmatrix}\begin{pmatrix} v_{n-1} \\ h_{n-1} \\ f_{n-1} \end{pmatrix} = \begin{pmatrix} v_n \\ h_n \\ f_n \end{pmatrix}$$

↓ one vector on both sides, no subscripts

$$\begin{pmatrix} 0.80 & 0.10 & 0.35 \\ 0.20 & 0.50 & 0.25 \\ 0 & 0.40 & 0.40 \end{pmatrix}\begin{pmatrix} v \\ h \\ f \end{pmatrix} = \lambda\begin{pmatrix} v \\ h \\ f \end{pmatrix}$$

In words. An eigenvalue equation says the matrix sends a vector to a number λ times that same vector. So the two different vectors become one vector, with no subscripts, and λ goes in front of it on the right.

Common mistakes

- Keeping the subscripts: an eigenvector is one vector, the same on both sides.
- Writing $= \lambda$ alone on the right: the right side must still be a vector.
- Putting λ on the side with the matrix: the matrix scales the vector by λ.

Problem 9 How many eigenvalues, and which one?

a. Three, and five: an $n \times n$ matrix has n eigenvalues, counting repeats.

b. Start from what "settled" means:

Settled numbers do not change from one year to the next.

↓ so the matrix sends the vector to itself

$$\boldsymbol{A}\vec{x} = 1\,\vec{x}$$

Why: $\lambda = 0.5$ would cut every number and the total in half each year, but the total is conserved.

In words. $\lambda = 1$ is the reasonable one: with it, the vector stays the same from year to year, so the numbers stay put and the total is kept. A λ below 1 would shrink the total.

Common mistakes

- "Because it is the biggest": say why, that the vector stays the same and the total is kept.
- "Because the deer stop moving": the deer keep moving; the numbers stay the same because the flows balance.

4. Verify by multiplying

Problem 10 Multiply a matrix by a vector

a. Each row times the whole vector:

$$2(3) + 1(1) + 0(2) = 6 + 1 + 0 = 7$$
$$1(3) + 3(1) + 2(2) = 3 + 3 + 4 = 10$$
$$0(3) + 4(1) + 1(2) = 0 + 4 + 2 = 6$$

↓ one result for each row

Conclusion

The result is $(7, 10, 6)$.

In words. Each entry of the result is one row of the matrix times the whole vector: first with first, second with second, third with third, then add.

b. $2 - 2 + 9 = 9$, $0 - 1 + 12 = 11$, and $4 + 0 + 3 = 7$, so the result is $(9, 11, 7)$.

c. $240 + 30 + 140 = 410$, $60 + 150 + 100 = 310$, and $0 + 120 + 160 = 280$, so the result is $(410, 310, 280)$. The total stays 1000.

Common mistakes

- Multiplying a column by the vector: in (a), $2(3) + 1(1) + 0(2)$ uses row 1, not column 1.
- Dropping a minus sign in (b): $2(-1) = -2$ and $1(-1) = -1$.
- Skipping the 0 entries in a way that shifts the others: each entry of a row still lines up with its own entry of the vector.

Problem 11 Verify an eigenvector

$$0.70(800) + 0.15(800) + 0.30(400) = 560 + 120 + 120 = 800$$
$$0.20(800) + 0.60(800) + 0.40(400) = 160 + 480 + 160 = 800$$
$$0.10(800) + 0.25(800) + 0.30(400) = 80 + 200 + 120 = 400$$

↓ compare with the vector

Conclusion

The result is $(800, 800, 400)$, the same vector, so $\lambda = 1$.

In words. Each entry of the result is one row of the matrix times the whole vector. The result is 1 times the vector we started with, so it is an eigenvector with $\lambda = 1$.

Common mistakes

- Multiplying a column by the vector: multiply each row, first with first, second with second, third with third.
- Stopping after one row: every row must match.

Problem 12 Multiply to find λ

$$0.80(-50) + 0.10(10) + 0.35(40) = -40 + 1 + 14 = -25$$
$$0.20(-50) + 0.50(10) + 0.25(40) = -10 + 5 + 10 = 5$$
$$0(-50) + 0.40(10) + 0.40(40) = 0 + 4 + 16 = 20$$

↓ compare entry by entry with $(-50, 10, 40)$

Conclusion

Every entry was multiplied by the same number: $\lambda = 0.5$.

In words. The result $(-25, 5, 20)$ is 0.5 times the vector, so $\lambda = 0.5$. Check more than one entry: all of them must give the same factor.

Common mistakes

- Losing a minus sign: $0.80(-50) = -40$.
- Answering with the result instead of λ: λ is the factor between the result and the vector.

Problem 13 Not every vector is an eigenvector

a. $700 + 90 + 120 = 910$, $200 + 360 + 160 = 720$, and $100 + 150 + 120 = 370$, so the result is $(910, 720, 370)$.

b. No. The first entry went from 1000 to 910, but the second went from 600 to 720: the result is not one number times the same vector.

c. Every vector keeps its total here, because each column adds up to 1. An eigenvector must also keep its direction: every entry multiplied by the same number.

Common mistakes

- Calling it an eigenvector because the total stayed the same: compare entry by entry.

5. Physical meaning

Problem 14 What the eigenvector means

The entries add up to 10

↓ scale so they add up to 1000

$$\frac{1000}{10} = 100, \quad \text{so } (500, 300, 200)$$

In words. After many years the herd settles at 500 deer in Valley, 300 in Hills, and 200 in Forest. Those numbers then stay the same from year to year, even though deer keep moving, because what leaves each habitat equals what arrives.

Common mistakes

- "The deer stop moving": the numbers stop changing, but the deer do not.
- "These are the numbers at the start": an eigenvector with $\lambda = 1$ is where the model ends up.

Problem 15 Still moving at equilibrium

a. Each rate is a fraction of the group it leaves. Valley to Hills: $0.20 \times 500 = 100$. Hills to Valley: $0.10 \times 300 = 30$. Hills to Forest: $0.40 \times 300 = 120$. Forest to Valley: $0.35 \times 200 = 70$. Forest to Hills: $0.25 \times 200 = 50$.

b. Valley: 100 leave, and $30 + 70 = 100$ arrive. Hills: $30 + 120 = 150$ leave, and $100 + 50 = 150$ arrive. Forest: $70 + 50 = 120$ leave, and 120 arrive.

c. No. Hundreds of deer move every year, but what leaves each habitat equals what arrives, so the numbers stay the same.

Common mistakes

- Taking the rate of the group the deer arrive at: Forest to Hills is 25% of Forest, not of Hills.